

C2-1

1. (a). $\int_{-\infty}^{+\infty} |\psi(x, t)|^2 dx = 1$, $\psi(x, t) = A e^{-\lambda|x|} e^{-i\omega t}$

$$\begin{aligned} \int_{-\infty}^{+\infty} |\psi(x, t)|^2 dx &= \int_{-\infty}^{+\infty} A^2 (e^{-\lambda|x|})^2 dx = 2A^2 \int_0^{+\infty} e^{-2\lambda x} dx \\ &= \left(-\frac{1}{2\lambda} e^{-2\lambda x} \times 2A^2 \right) \Big|_0^{+\infty} \\ &= \frac{A^2}{\lambda} = 1 \end{aligned}$$

则 $A = \sqrt{\lambda}$

(b) $\langle x \rangle = \langle \psi | x | \psi \rangle$

$$= \int_{-\infty}^{+\infty} \psi^*(x, t) \cdot x \psi(x, t) dx = \int_{-\infty}^{+\infty} A^2 e^{-2\lambda|x|} \cdot x dx$$

$$= \lambda \cdot \int_{-\infty}^{+\infty} e^{-2\lambda|x|} x dx.$$

$g(x) = x e^{-2\lambda|x|}$ $g(x) = -g(-x)$, 则 $g(x)$ 奇函数, $\int_{-\infty}^{+\infty} g(x) dx = 0$

故 $\langle x \rangle = 0$

$\langle x^2 \rangle = \langle \psi | x^2 | \psi \rangle$

$$= \lambda \int_{-\infty}^{+\infty} e^{-2\lambda|x|} x^2 dx.$$

$$= 2\lambda \int_0^{+\infty} e^{-2\lambda x} x^2 dx = - \int_0^{+\infty} x^2 d(e^{-2\lambda x})$$

$$= - \left[x^2 e^{-2\lambda x} \Big|_0^{+\infty} - \int_0^{+\infty} e^{-2\lambda x} \cdot 2x dx \right]$$

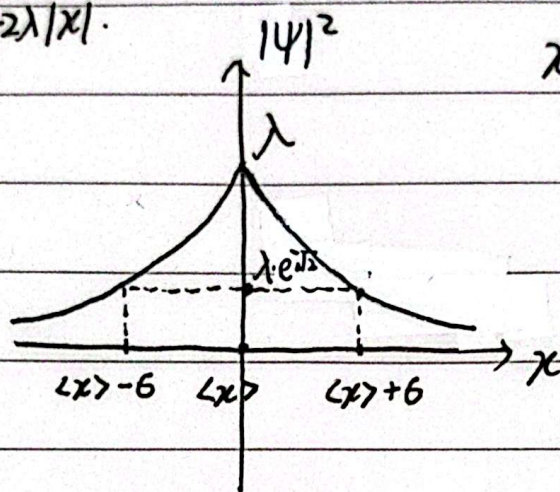
$$= \int_0^{+\infty} e^{-2\lambda x} \cdot 2x dx = \frac{1}{2} \left(-\frac{1}{\lambda} 2x e^{-2\lambda x} - \frac{1}{\lambda^2} e^{-2\lambda x} \right) \Big|_0^{+\infty}$$

$$= \frac{1}{2\lambda^2}$$

故 $\langle x \rangle = 0$, $\langle x^2 \rangle = \frac{1}{2\lambda^2}$

(c) 标准差 $\sigma = \sqrt{\langle x^2 \rangle - \langle x \rangle^2} = \frac{\sqrt{2}}{2} \cdot \frac{1}{\lambda}$

$|\psi|^2 = \lambda \cdot e^{-2\lambda|x|}$



$x = \langle x \rangle + \sigma = 0$ 时

$|\psi|^2 = \lambda \cdot e^{-2\lambda \cdot \frac{\sqrt{2}}{2} \cdot \frac{1}{\lambda}} = \lambda \cdot e^{-\sqrt{2}}$

偶函数, 则 $x = -\sigma$ 时,

$|\psi|^2 = \lambda \cdot e^{-\sqrt{2}}$

(d) $P = 2 \int_{\sigma}^{+\infty} |\psi|^2 dx = -e^{-2\lambda x} \Big|_{\sigma}^{+\infty} = e^{-2\lambda \sigma} = e^{-2\lambda \cdot \frac{\sqrt{2}}{2} \cdot \frac{1}{\lambda}} = e^{-\sqrt{2}} = 0.2431$

2. (a) $\psi_c = \psi_a \cdot \cos \alpha + \psi_b \cdot \sin \alpha$

(b) $P = |\cos \alpha|^2$

(c) $\langle \psi_c | \psi_c \rangle = (\langle \psi_a | \cos \alpha + \langle \psi_b | \sin \alpha) \cdot (|\psi_a \rangle \cos \alpha + |\psi_b \rangle \sin \alpha)$

由于 $\langle \psi_a | \psi_a \rangle = 1$, $\langle \psi_b | \psi_b \rangle = 1$, $\langle \psi_a | \psi_b \rangle = 0$, $\langle \psi_b | \psi_a \rangle = 0$

则 $\langle \psi_c | \psi_c \rangle = \langle \psi_a | \psi_a \rangle \cdot \cos^2 \alpha + \langle \psi_b | \psi_b \rangle \cdot \sin^2 \alpha$

$+ (\langle \psi_a | \psi_b \rangle + \langle \psi_b | \psi_a \rangle) \cdot \sin \alpha \cos \alpha$
 $= \cos^2 \alpha + \sin^2 \alpha = 1$

故归一化.